



Article

Structural and Fractal Analysis of Soil Cracks Due to the Roots of *Setaria Viridis*

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Abstract: Soil surfaces form complex crack networks as a result of water loss and shrinkage. A crack network destroys the integrity of the soil and becomes the main factor affecting rainfall infiltration, slope instability and soil integrity. In this paper, a soil fracture network is quantified using fractal characteristics and fractal dimensions, and the soil fracture network is identified and calculated using digital image processing technology. The fracture network of silty clay with different setaria viridis root content is studied during the process of evaporation. Saturated mud is prepared by taking soil samples and collecting setaria viridis roots. The content of setaria viridis roots in each saturated mud sample is 0 g, 0.1 g, 0.2 g and 0.3 g. In the artificial climate environment simulation system, thin-layer root soil is dried by controlling the temperature and humidity to simulate dry climate conditions. During the test, the crack development process is recorded using a digital camera. The results show that when the root content is 0, 0.1 g, 0.2 g and 0.3 g, the water content values when a fissure is generated are 35.2%, 41.2%, 42% and 46.4%, and the initial fractal dimension values are 1.100, 1.106, 1.112 and 1.115, respectively. The fractal dimension value increases rapidly in the early stage of fissure generation, and it reaches the maximum value when the water content reaches 13.66%, 15.2%, 15.66% and 17.98%, respectively. According to the change law of the fractal dimension, the fractal dimension increases rapidly following the initial appearance of the fracture, and, with a continuous reduction in water content in the later stage, the fracture characteristics gradually stabilize, and the change rate of the fractal dimension becomes slow.



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Keywords: fractal analysis; image processing; soil crack network; fracture rate; setaria viridis root system

1. Introduction

In nature, everything from a single atom to the whole Earth and even the universe is composed of matter, and various substances have a wide range of nonlinear distribution. Soil is a porous medium composed of solid particles and pores of different sizes and shapes. It is a loose material composed of various mineral particles with various mechanical properties. Soil is prone to cracking, with the cracks having different shapes, lengths and widths, due to changes in the external environment. These cracks have obvious nonlinearity and self-similarity [1]. This nonlinear characteristic is the basic characteristic of fractal research. Crisscross network fissures formed on a soil surface [2–5] have an important impact on the engineering properties of the soil and, thus, lead to various engineering geological problems and disasters. For example, cracks greatly weaken the bearing capacity of the soil [6–8], damage the integrity of the soil structure [9] and reduce the strength of the soil, but, at the same time, cracks also provide a convenient channel for rainwater to quickly infiltrate into the soil [10–12]; they also exacerbate slope instability and induce landslides, mudslides and other disasters [13–15]. The existence of cracks increases the weathering depth of the soil mass, aggravates soil erosion on the slope surface and damages

the ecological environment [16–19]. It can be seen that the influence range of soil fissures is extremely wide, involving many disciplines, such as civil engineering, geology, water conservancy and environmental engineering.

The root system is one of the most important organs used by plants to adapt to land life for a long period of time. It is composed of the main roots and lateral roots. It plays a role in fixing and supporting plants, and it absorbs water and nutrients through the transport of root microtubules. It plays an important role in the biogeochemical cycle of the ecosystem [20]. Root growth activity is a quantity that represents the development degree of plant roots. The root growth activity of plants directly affects the nutritional status and yield level of aboveground plants [21]. The root system produces secretion to the outside during the growth process. The high-molecular viscous polysaccharide produced by secretion has strong adhesion to soil particles [22], which has a significant impact on the physical properties, such as the stability, size and distribution of soil microaggregates, thus improving the cohesion of the soil. However, the root structure of plants is regulated by the root biological clock, which further affects the interaction between roots and soil [23]. The plant root system improves the physical and chemical properties of the soil via interlacing, intercalating, network consolidating and root–soil bonding in the soil [24], greatly improving the soil's erosion resistance, improving the water and soil conservation capacity and reducing the desertification and loss of the surface soil. The existence of a root system in soil can play the role of reinforcement and anchorage, so the plant root system is widely used in slope protection. At the same time, the existence of a plant root system can also play the role of stabilizing the water content of the surface soil and controlling soil cracks [25,26].

Employing a plant root system in topsoil is a good engineering solution to improve the soil structure and soil structure stability [27,28]. As is well known, soil moisture content is a major factor affecting soil strength. On the one hand, soil moisture is lost through surface evaporation, and on the other hand, the water in the soil is absorbed by the root system and reduced by page evaporation, which can lead to an increase in soil suction. This increase in suction leads to an increase in soil strength [29]. Since soil and root systems are two different characteristic materials and there are numerous pile root systems distributed on the main root system, the existence of these root systems plays a role in reinforcement, anchorage and support, enhancing the tensile and shear strengths of the soil, as well as controlling the size and degree of soil cracking. Root systems serve the purpose of strengthening the soil mass and stabilizing slopes [30–35]. On the one hand, the reinforcement effect of a root system on soil is related to the degree of interaction between the root system and the soil, and it is closely related to the stress and deformation mode of the root system and the physical and mechanical properties of the soil. On the other hand, the existence of a root system causes the water in the soil to dissipate rapidly, thus causing the pore water pressure to dissipate. At the same time, the existence of a root system destroys the link with the soil formed by the suction between the particles, forming a two-phase structure, increasing the water discharge channel, accelerating the evaporation rate of the soil water and enhancing the adsorption force between the particles in the rapid evaporation area of the soil, thereby forming cracks in the suction concentration area [36,37]. Against the background of global climate change, extreme droughts and high-temperature weather have frequently occurred in recent years. The problem of soil shrinkage cracking with plant roots has increasingly attracted scholars' attention.

Fractal theory was introduced in the mid-1970s. It is a general term for irregular structures and configurations with self-similarity. The characteristics of this structure are that the local and the whole are similar in some way. Its research object is a system with self-similarity due to the disorder (irregular) widely existing in nature and social activities [38]. Due to the internal and external stresses of the Earth, rock and soil mass, various criss-cross fracture networks form (faults, joints, bedding, schistosity, etc.), which become the migration channels and storage sites of oil, natural gas, gas and radioactive substances [39]. The spatial distribution of fracture networks in a rock mass has self-similarity at different scales. This provides an application domain for fractal theory to quantify rock fracture

networks [40]. In engineering construction, due to differences in the water binder ratio and load, concrete is subject to changes in external conditions. The self-similarity of dynamic evolution processes, such as crack distribution, crack propagation and concrete damage, provides good conditions for studying the morphological characteristics of cracks using fractal theory [41]. As the soil mass for engineering construction, it is a typical porous medium aggregate. The clay particles in the soil are prone to producing shrinkage cracks under the condition of water loss. These cracks are affected by external conditions (such as temperature, humidity, root system, etc.) and produce cracks with different widths, lengths and depths. The pore structure is extremely complex. The distribution of these cracks in the soil mass has obvious nonlinear self-similarity characteristics [42], which conform to fractal characteristics; fractal theory can be used to comprehensively evaluate the complexity of soil cracks.

In this paper, using plant roots, silty clay is slurried, mixed evenly according to the root content and tested under the same evaporation environment. Then, the crack expansion and evolution law of the slurry soil with different root content is analyzed during the evaporation process. Fractal theory is used to evaluate the complexity of the soil cracks under different plant roots, and the evaporation rate, fracture rate, average fracture width and fracture propagation characteristics are analyzed.

2. Materials and Methods

2.1. Research Background

Henan Province is located in the middle east of China, the middle and lower reaches of the Yellow River and the hinterland of Eurasia. Henan Province is high in the west and low in the east. The Taihang mountain range, the Funiu mountain range, the Tongbai mountain range and the Dabie mountain range are in the north, west and south, and they are distributed in a semi-circular pattern along the provincial boundary. The central and eastern parts form the Huang Huai Hai Plain. The southwest is the Nanyang basin. Xuchang City is located in the central part of Henan Province, China, bordering Zhengzhou in the north. The Funiu mountain range and the Zhongyue Songshan mountain are located in the west. The Huanghuai sea plain is located in the east and south, between $112^{\circ}42' - 114^{\circ}14'$ E and $34^{\circ}16' - 34^{\circ}58'$ N. It belongs to the transition zone from the remaining veins of the Funiu mountain to the eastern Henan Plain. The terrain slopes from west to east. In the west, it is the middle and low mountain hilly area of the rest of Funiu mountain, where the highest elevation is 1150.6 m. The central part is a hilly area formed by the slow rise of the basement structure and denudation. The eastern part is the Huanghuai alluvial plain, with a minimum elevation of 50.4 m. Moreover, 75% of the area in the territory is plain, 25% of the area is hills, and the average annual total water resources amount to 510 million cubic meters. It belongs to a warm, temperate, sub-humid monsoon climate. The annual average temperature is around 15°C , the average temperature in January is 0.7°C , the average temperature in July is 27.1°C , the average annual sunshine amounts to 2280 h, and the annual precipitation is approximately 700 mm.

Silty clay is one of the main strata in the Xuchang area, and it has the properties of water loss and shrinkage. The annual precipitation is low and extremely uneven, and it is characterized by more rain in summer and autumn and less rain in winter and spring. Therefore, the surface often appears to have dry cracks, the water loss rate is accelerated, and the slope shrinks and cracks, which make agricultural water conservation and engineering slope stability difficult. Therefore, it is of great significance to study plant roots for soil water control.

2.2. Materials

The setaria viridis roots selected in the experiment came from setaria viridis (Figure 1), which is an annual herb. The roots are bearded, and tall setaria viridis plants have supporting roots. The seedling height is around 15 cm, and the leaf surface is 4–6 cm. The root system is mainly composed of 4–6 taproots. The taproot is extensive and long, with a

length of 6–8 cm and a diameter of 0.5–1 mm. The taproot is attached to the fibrous root, which is 1–2 cm long and can be well attached to the soil.

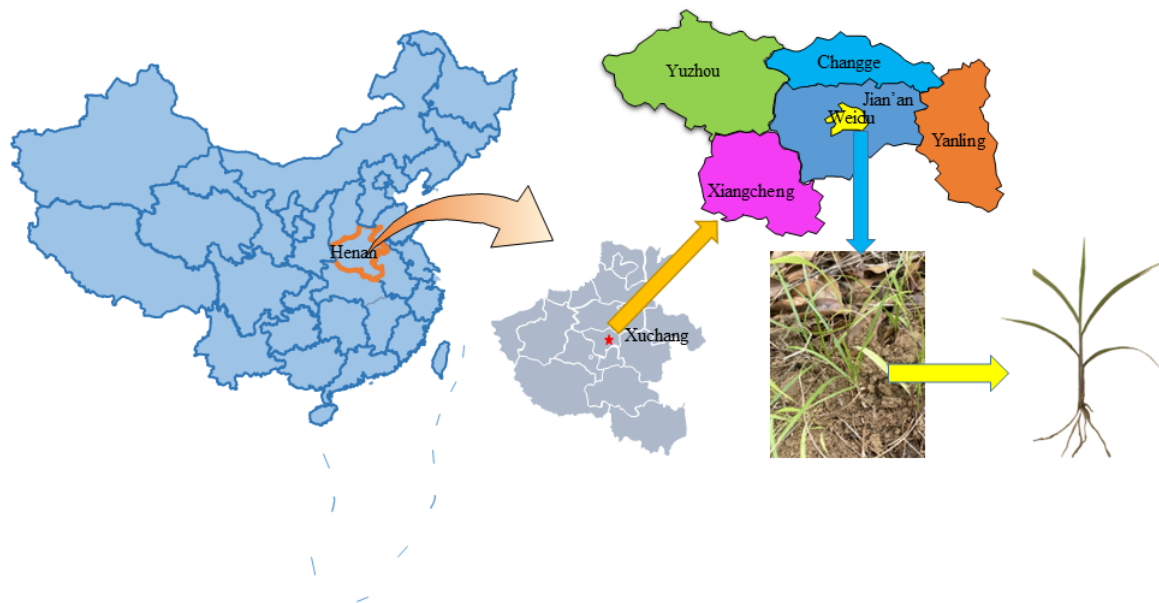


Figure 1. Study area and sampling location.

The test soil was collected from cultivated surface soil in Xuchang City, Henan Province. The color of the soil was reddish brown. The sampling position is shown in Figure 1.

The soil was crushed after air drying to remove *setaria viridis* roots, gravel and other sundries, and then it was filtered with a 2 mm sieve. The Atterberg limit (limit moisture content), density and moisture content of the undisturbed soil were determined according to the national standard for test methods of geotechnical engineering (GB/T 50123-2019). Table 1 presents the physical properties of the soil samples. Using a laser particle size analyzer (bettersize 2000; measurement range: 0.02–2000 μm), a particle size analysis test was carried out on the soil sample, and the measured particle size distribution curve is shown in Figure 2. According to the engineering classification of soil in the national standard for test methods of geotechnical engineering, the test soil was low-liquid-limit clay.

Table 1. Physical properties of test soil.

Specific Gravity	Natural Water Content /%	Density /(g/cm^3)	Plastic Limit %	Liquid Limit %	Plasticity Index
2.72	9.3	1.76	20.4	35.4	15

2.3. Experimental Process

The 2 mm screened soil was placed on a tray and inserted into a blast drying oven. The temperature was adjusted to 105 $^{\circ}\text{C}$, and the drying time was 12 h. After this, the dried soil sample (250 g for each sample) was placed in an open plexiglass container with a length, width and height of 245 mm $\times$ 245 mm $\times$ 40 mm, respectively. In order to facilitate the full mixing of the *setaria viridis* roots and the mud soil, the collected grass roots were cut off and placed together for natural air drying. The collected roots were placed into plexiglass containers according to weights of 0 g, 0.1 g, 0.2 g and 0.3 g, and they were fully mixed with the soil to prepare saturated mud samples with an initial water content of 100%. The plexiglass container was placed in a sealed plastic bag for 24 h to stabilize the mud deposition. Then, the samples were placed into the artificial climate environment simulation system (ZHS-030; internal space size: W $\times$ L $\times$ H = 3500 $\times$ 4400 $\times$ 2000 mm), where the temperature and relative humidity were set to 30.5 $^{\circ}\text{C}$ and 60%, respectively. Four groups of samples with different root content were prepared in this experiment. To

ensure the reliability of the test data, three parallel samples were prepared for each group of samples. The group of samples with a root content of 0 was taken as the control sample.

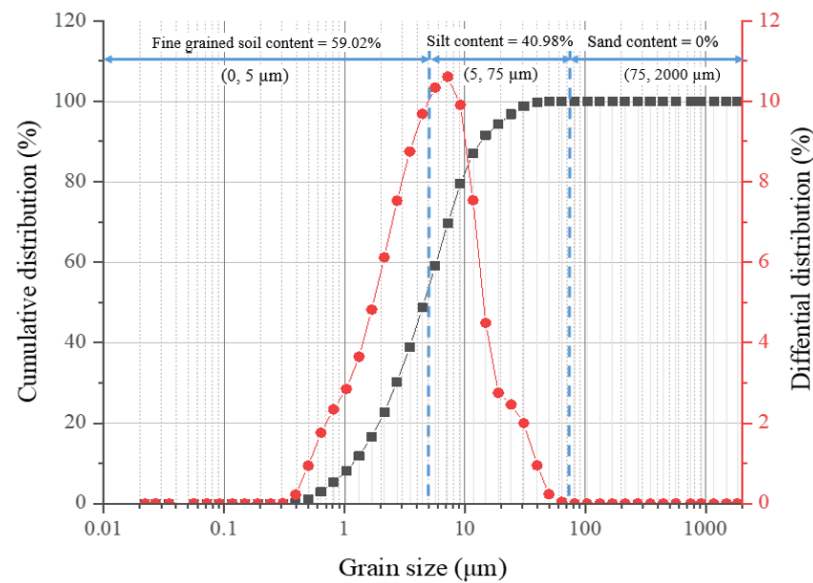


Figure 2. Particle distribution curve of test soil.

A weighing and photographing device was designed, as shown in Figure 3. The samples were measured using an electronic analytical balance with accuracy of 0.01 g, and the change in sample mass was recorded. At the same time, a digital camera (Canon Eos850d, 24.1 million pixels) was used to record the evolution process of surface cracks. In order to ensure that the distance and size of the pictures taken were the same, the digital camera was fixed on a steel bracket, and the lens was perpendicular to the center of the sample and kept at a fixed distance from the sample. The light intensity in the climate environment simulation system was adjusted according to the needs, and the light intensity was kept unchanged during the test process to obtain the same high-quality crack pictures. The quality and image data of each sample were collected every hour and stored on a computer. When the soil quality difference measured two consecutive times was less than 0.1 g, the evaporation process was considered to be completed, and the test was stopped. The water content w and evaporation rate E_r of the sample during drying were calculated using Formulas (1) and (2), respectively.

$$w = \frac{(m_0 - m_s - m_d) - (m_0 - m_t)}{m_s} = \frac{m_t - m_s - m_d}{m_s} \quad (1)$$

$$E_r = \frac{\Delta m}{\Delta t} \quad (2)$$

where m_s : quality of dry soil; m_0 : total mass of soil sample and soil sample box at the initial time of sample; m_t : total mass of soil sample and soil sample box at time t of the evaporation process; m_d : mass of the container; and Δm and Δt are the mass difference and time interval of two adjacent weights of the sample, respectively.

2.4. Image Processing

In order to evaluate the influence of different root content on soil evaporation and fissure evolution, the images obtained during the experiment were processed to analyze the fissure characteristics in the images. The image processing process is shown in Figure 4. First, the sample area of the soil was extracted from the original test photos, and the sizes of the captured images were uniformly adjusted to 3100×3100 pixels in order to ensure that the image size after the clipping of each sample was exactly the same (Figure 4a). In

the second step, due to the obvious difference between the brightness of the crack and the background, in order to better reflect the characteristics of the crack, the color RGB image was converted into a gray-scale image (Figure 4b). In the third step, the gray-scale image was converted into a binary image by using the global threshold method. The black area represents the fissure, and the white area represents the soil (Figure 4c). In the fourth step, morphological opening and closing operations were used to remove noise, fill holes, bridge cracks, etc. (Figure 4d). In the fifth step, the fracture skeleton of the denoised binary image was extracted to obtain a fracture network with a width of 1 pixel. The burr on the bone, which was the noise generated during skeletonization, was removed (Figure 4e) [43,44].

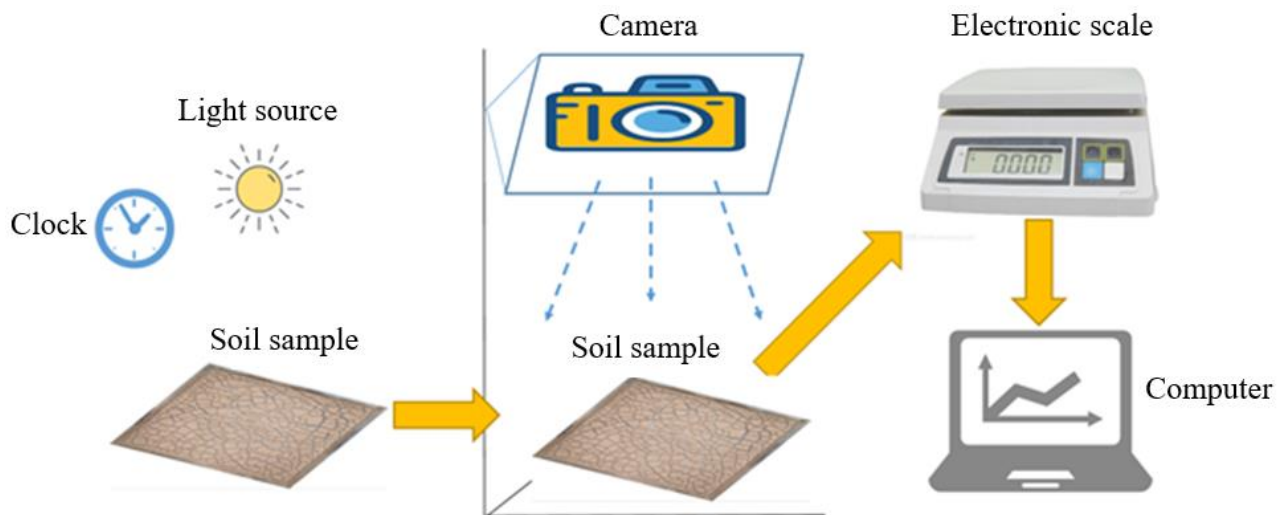


Figure 3. Schematic diagram of weighing and photographing device.

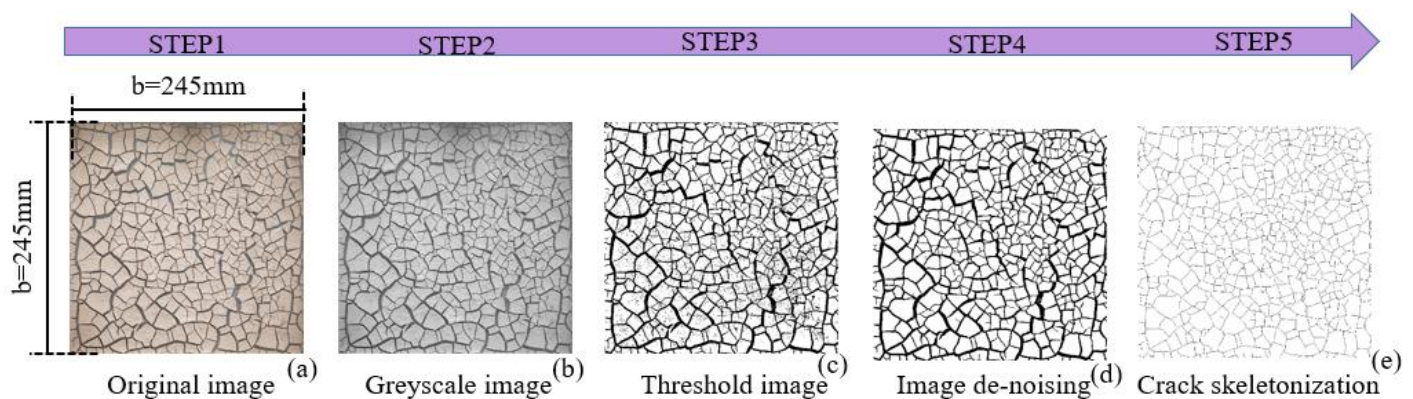


Figure 4. Image processing steps.

The crack rate was defined as the ratio of the number of black pixels to the total number of pixels in the denoised binary image. Fractal theory was used to further describe the irregularity and complexity of the fracture network. The box dimension was used to calculate the fractal dimension, and the minimum number of grids covered the whole fracture area $N(k) = 1/k^2$. The $\lg N(k) - \lg(k)$ double logarithmic coordinate plane was used for data processing in the fracture map, and the fractal dimension D could be calculated by calculating the slope of the fitting straight line [45]. The calculation formula for the fractal dimension is

$$D = -\lim_{k \rightarrow 0} \lg N(k) / \lg(k) \quad (3)$$

In the calculation formula, D is the fractal dimension; k is the side length of the grid; and $N(k)$ is the number of meshes.

3. Analysis of Test Results

3.1. Effect of Root Distribution on Fracture Distribution

Figure 5 shows the fracture images without a root system at four evaporation time points (56 h, 62 h, 72 h and 106 h). When there is no root distribution in the soil, because the soil and the plexiglass container box belong to different materials, the cracks preferentially form from the boundary of the box soil and gradually develop to the center. The crack formed at the edge is affected by the boundary crack, and the soil shrinks toward the center to form a unidirectional shrinkage crack. The width of the crack formed is small. When the soil expands to the center, the soil first cracks at the position with the lowest tensile strength. With continuous evaporation, the fractures continue to expand along the tip, and finally the fractures are connected with each other to form a connected fracture network. At this time, the fracture width formed is large and relatively uniform. The length of a single main fracture is long. The larger the fracture width, the larger the cut block formed (Figure 5d).

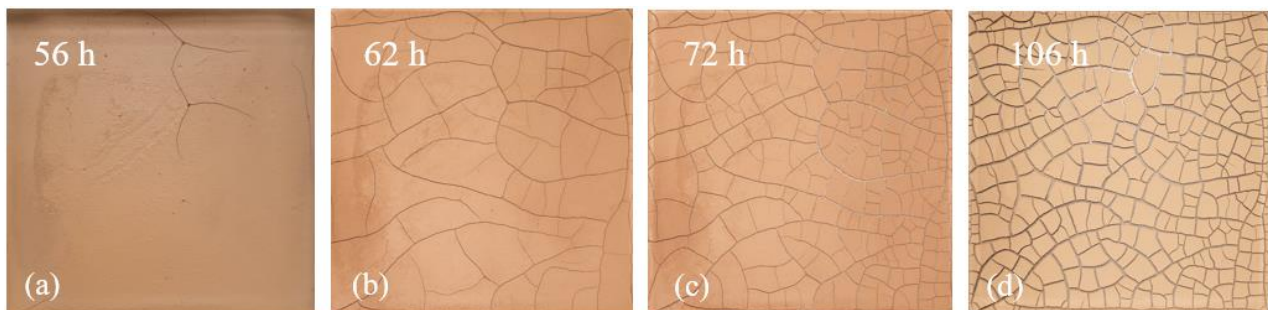


Figure 5. Characteristics of soil fissure development without root system. (a) Cracks appear; (b) Main fissure formation; (c) Secondary fissure formation; (d) Fissure extension.

Figure 6 shows the development characteristics of soil fissures with root systems at four evaporation time points (56 h, 62 h, 74 h and 110 h). Under the influence of the root system, the soil drying shrinkage cracks show greater heterogeneity. When a certain root system is distributed in the soil (Figure 6) and the soil moisture content is 54%, the soil first cracks along the root distribution direction (Figure 6e) and forms major cracks (Figure 6f). The existence of cracks expands the contact area between the soil and the external environment, so the water content at the main crack position decreases rapidly, the crack extension speed accelerates, the crack width increases, and the secondary crack expansion speed around the main crack also rapidly increases (Figure 6g,h). The morphological characteristics of the fracture network under the action of the root system are obviously different near the root system and far from the root system. The degree of fracture cracking near the root system is small, the width of a single fracture is small, and the area of the cutting body formed is small and the number is large. The crack at the position far from the root system is relatively large, the length of a single crack is relatively wide and long, and the area of the cutting body is relatively large, which is basically consistent with the shape of the soil dry shrinkage crack under the action of no root system.

When following the root system of *setaria viridis*, the difference between the root system and the soil forms a weak area on the soil surface, and a tensile crack preferentially forms at this position. In the direction of a vertical root system, the existence of a root system limits the expansion of cracks. With a decrease in the water content, the *setaria viridis* roots play an important role in controlling the crack propagation characteristics (Figure 7). On the one hand, the root system promotes the formation and expansion of cracks, cutting and fragmenting the soil; on the other hand, the existence of *setaria viridis* roots controls soil shrinkage. The moisture content of the soil changes from a saturated state to an unsaturated state. According to the mechanics theory of unsaturated soil, with the continuous evaporation of water, the moisture content gradually decreases, the matrix

suction between the soil particles gradually increases, and the main fracture is preferentially formed in the stress concentration area. When the main fracture extends along the tip to the crop root reinforcement area, it will encounter greater resistance. At this time, the fracture is forced to change direction and expand along the edge of the root reinforcement area. In the densely distributed area of the root system, a large number of root systems exist. When the matrix suction of the soil does not exceed the strength of the root system, the strength of the root system can well inhibit the tensile action formed by the matrix suction, and the integrity of the soil is better (Figure 7a) [46]. When the root system is sparsely distributed and the matrix suction is weak, the root system can control the expansion of the crack. When the tensile stress formed by the suction exceeds the root system strength, a dry shrinkage crack finally appears at the vertical root system position. At this time, the root system cannot effectively control the expansion of the crack (Figure 7b). When there is no setaria viridis root system, the crack expansion degree is large, and most of the cracks formed in the initial stage become main cracks (Figure 7c). According to the distribution of the root system and the expansion of the cracks, the higher the content of the root system in the soil, the more uneven the distribution of the crack width. The main crack is thinner at one end, and the secondary crack is smaller. When there is no setaria viridis root system, the width of the main fissure is relatively consistent, and the width of the secondary fissure is also relatively consistent.

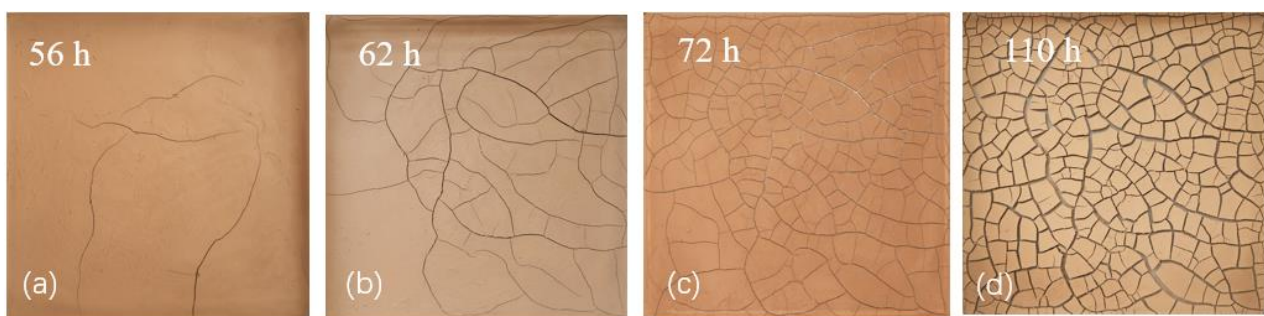


Figure 6. Development characteristics of soil fissure with root system. (a) Cracks appear; (b) Main fissure formation; (c) Secondary fissure formation; (d) Fissure extension.

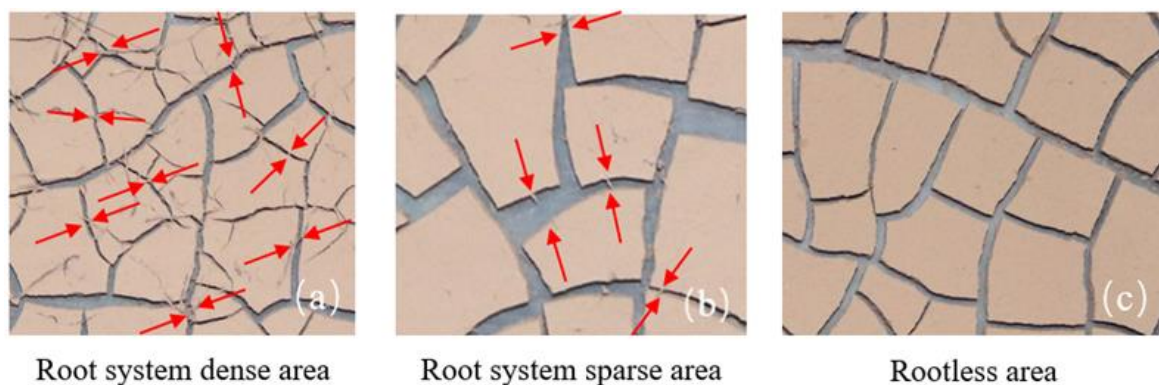


Figure 7. Distribution characteristics of setaria viridis root cracks.

3.2. Effect of Root Content on Fractal Dimension of Soil Fissure Network

Figure 8 displays a graph showing the relationship between the fractal dimensions of the soil fractures and the water content. After the initial appearance of fractures, the fractal dimension increases rapidly, and the change rate of the fractal dimension is large. With a further reduction in the water content, the fractal dimension further increases, but the increase rate obviously decreases. When there is no root system, the soil fracture network begins to appear when the water content is 35.2%, and the fractal dimension is 1.1. When the water content is reduced to 33.8%, the fractal dimension value rapidly increases to 1.508,

accounting for 55.1% of the total change rate. Then, the change rate gradually decreases. After the water content is reduced to 13.66%, the fractal dimension reaches the maximum value of 1.84. When the root content in the soil is 0.1 g, when the water content is 41.2%, the fracture begins to occur. At this time, the fractal dimension value is 1.106. When the water content decreases to 39.4%, the fractal dimension value rapidly increases to 1.339, accounting for 33.2% of the total change rate. When the water content decreases to 15.2%, the fractal dimension reaches the maximum value of 1.807. When the root content in the soil is 0.2 g and the water content is 42%, a soil fracture network begins to appear. When the water content is reduced to 40.4%, the fractal dimension rapidly increases from the initial value of 1.112 to 1.400, accounting for 41.4% of the total change rate. When the water content is reduced to 15.66%, the fractal dimension reaches the maximum value of 1.807. When the root content in the soil is 0.3 g, and the water content is 46.4%, a soil fracture network begins to appear, and the fractal dimension value is 1.115. When the water content decreases to 43.2%, the fractal dimension value increases to 1.430, accounting for 45.7% of the total change rate. When the water content decreases to 17.98%, the fractal dimension reaches the maximum value of 1.805. According to the change law of the fractal dimension, the fractal dimension changes rapidly following the initial appearance of a fracture, and the change rate of the fractal dimension gradually decreases with a continuous reduction in the water content in the later stage. According to the distribution characteristics of the fractal dimension, the higher the content of the root system, the lower the change rate of the initial fractal dimension.

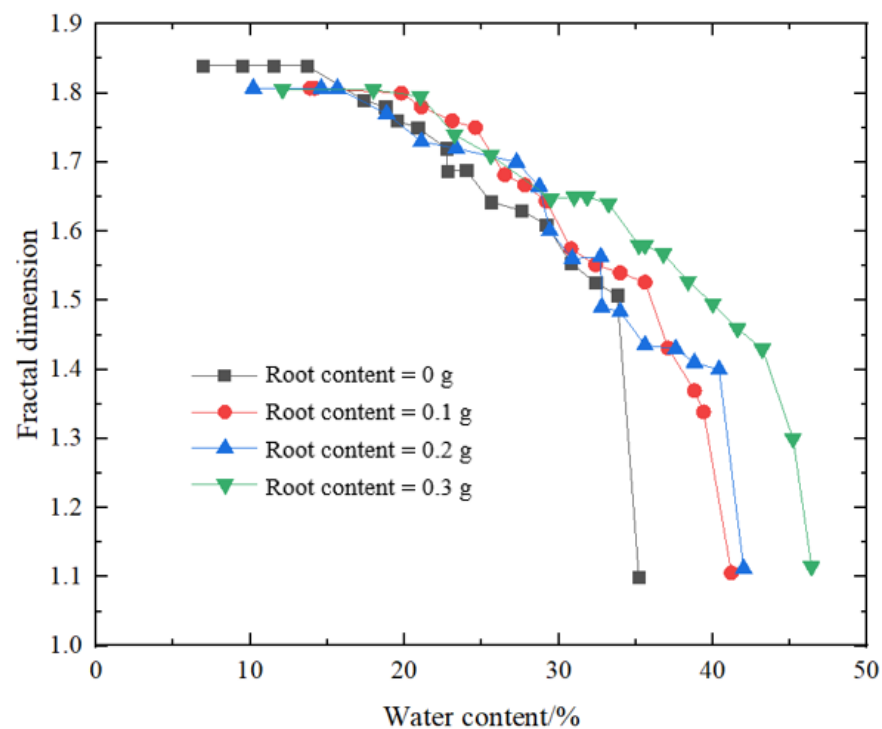


Figure 8. Relationship between fractal dimension of soil fissure and water content.

3.3. Effect of *Setaria Viridis* Root Content on Crack Rate and Crack Width

Figure 9a presents the variation curve of the soil fracture rate with water content. It can be seen in the figure that when there is no *setaria viridis* root system (root content = 0.0 g), cracks begin to appear when the water content is 35.2%. With a continuous decrease in the water content, the crack rate increases rapidly and tends to be stable after the water content decreases to 13.656%. When the root content is 0.1 g (root content = 0.1 g), after the water content decreases to 13.1%, the fracture rate reaches 19.58%, and the change is small. When the root content is 0.2 g (root content = 0.2 g), after the water content decreases to 11.2%, the change rate is small when the fracture rate reaches 19.21%. When the root content is 0.3 g

(root content = 0.3 g), after the water content is reduced to 10.9%, the change rate is gentle when the fracture rate reaches 18.85%. The overall situation is that with the increase in root content, the water content reaching the stable fracture rate is lower. From the change in the average crack width with the water content (Figure 9b), the greater the root content, the smaller the average crack width, which also reflects the influence of the root system on the crack size.

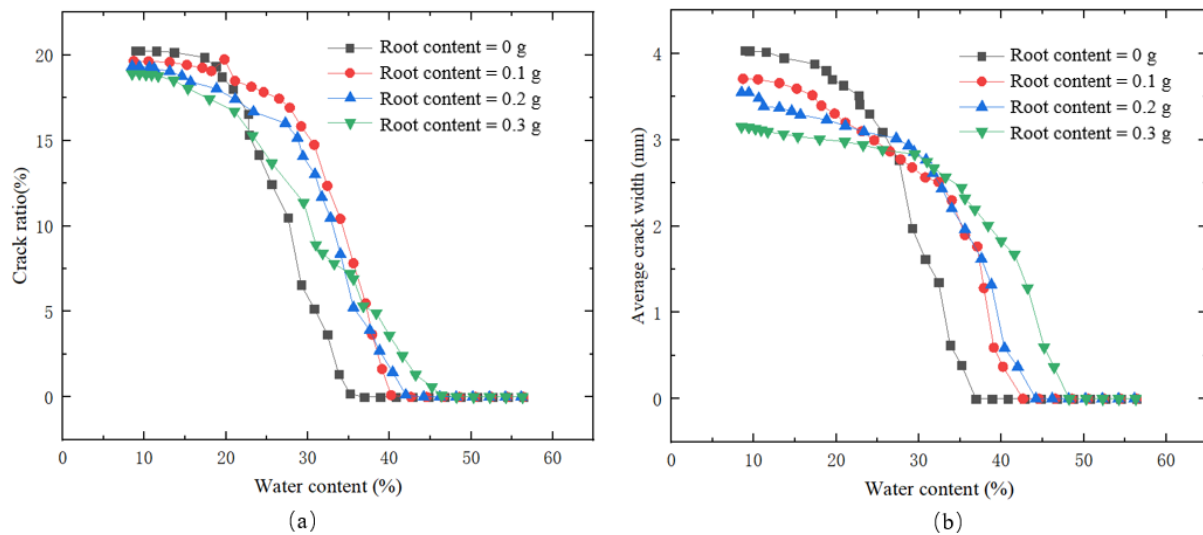


Figure 9. (a) Variation in soil fissure rate with water content and (b) variation in average width of soil fissure with water content.

4. Discussion

According to the evolution process of the surface cracks of the soil with a root content of 0.3 g during evaporation (Figure 10a–f), cracks begin to appear when the water content of the saturated soil changes to 46.2% after 56 h (Figure 10a); moreover, the water content changes to 41.6% after 62 h (Figure 10b), and the cracks in the soil sample increase significantly (Figure 10b). After 76 h, the water content of the soil sample changes to 31.84%, at which time cracks basically form in the soil sample (Figure 10d). In the later stage, the cracks continuously widen at 82 h and 110 h (Figure 10e–f). Figure 10g shows the relationship between the evaporation rate and the drying time. All samples were completely saturated at the beginning. In the process of evaporation drying, it can be divided into four areas. The early stage is the stable evaporation rate area, and the change trend of the water content of each sample is basically the same. When the *setaria viridis* root content is 0.0 g, 0.1 g, 0.2 g and 0.3 g, respectively, the water content of the soil that begins to crack is 35.2%, 41.2%, 42% and 46.2%, respectively. With the increase in time, cracks are generated, the evaporation rate of water content is accelerated, and it enters the evaporation enhancement area. During this period, the crack propagation rate increases, and the number of cracks also sharply increases until the cracks completely form. When the *setaria viridis* root content is 0.0 g, 0.1 g, 0.2 g and 0.3 g, respectively, the maximum evaporation reaches 2.5 g/h, 2.7 g/h, 2.75 g and 3.0 g/h. With the further increase in soil evaporation time, the length and width of cracks have been established, the water content of the soil has been further reduced, and the evaporation rate of the soil has entered the recession area. The figure shows that the higher the content of *setaria viridis* roots, the faster the decay rate. In the residual evaporation stage, the moisture content of the soil is nearly the same as that under the same external environment, and the moisture content does not change.

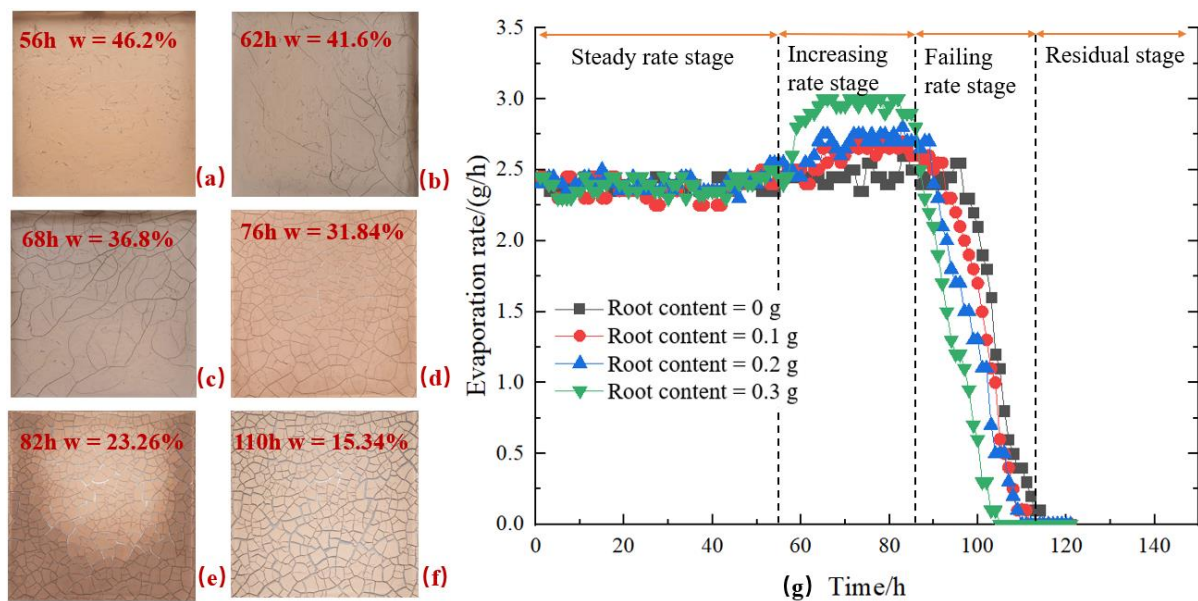


Figure 10. (a–f) Development process of surface fissure with root content of 0.3 g over time and (g) change in evaporation rate with time under different root content.

According to the evaporation process, when the water content is high in the early stage, the evaporation rate remains similar to that of surface evaporation. When the soil cracks, the contact area with the outside increases, and the evaporation rate increases. When the soil contains *setaria viridis*, the roots and stems of *setaria viridis* act as a form of natural cellulose. The evaporation of the leaves of *setaria viridis* causes the internal water of *setaria viridis* to decrease, and the evaporation of the leaves forms negative pressure inside, and then absorbs the water in the soil through the capillary roots of the roots. This also causes the evaporation process to be composed of two forms of evaporation: the evaporation of the soil surface and the evaporation of the *setaria viridis* roots when the soil contains them. Thus, the evaporation rate increases with an increase in the *setaria viridis* root content (Figure 11).

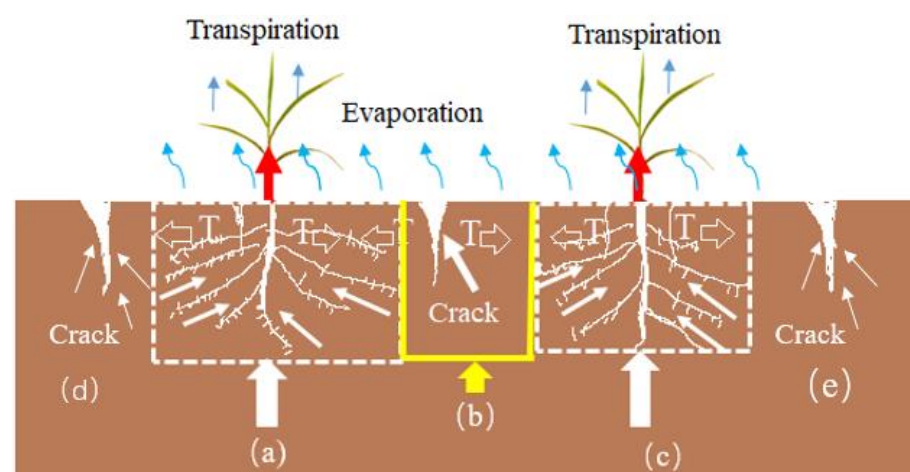


Figure 11. Mechanism of root soil crack. (a,b) Root distribution area; (c–e) Non root distribution area.

In the process of soil evaporation, when the soil moisture content is less than the shrinkage limit moisture content, the soil loses water and shrinks, resulting in tensile stress. In the non-root distribution area, when the tensile stress exceeds the binding force between the soil particles, cracks form (Figure 11b,d,e). At this time, the sizes of the main cracks formed are basically the same, and the sizes of the cutting blocks of the secondary cracks

are relatively uniform (Figure 7c) [47]. When there are setaria viridis roots in the soil, the setaria viridis roots and soil form a composite soil (Figure 11a,c), causing the friction and bonding strength of the root and soil interface to increase, and the existence of the roots accelerates the evaporation of soil moisture. According to the theory of unsaturated soil mechanics, this rapidly decreasing soil water content causes the matrix suction between soil particles to increase and the tensile tension to increase. At this time, the existence of the root system plays an anchoring role. When the tensile stress between the soil particles exceeds the strength of the setaria viridis root system, the root system can be broken to form cracks (Figure 11). At this time, the anchoring role of the root system is also involved in inhibiting the formation and expansion of cracks. The developed size of the cracks is related to the root strength.

5. Conclusions

1. Soil is a type of porous medium that is easily affected by changes in the external environment, and it easily produces water loss, shrinkage and cracking. As a type of water discharge channel and soil-reinforced fibrous tissue, plant root systems play important roles in soil cracking, which has also become an important means for vegetation to protect soil water and prevent slope cracking.

2. The setaria viridis root system plays an important role in controlling soil cracking and soil erosion. In this paper, setaria viridis roots and silty clay were used for slurring, and the evolution characteristics of fissure expansion during the process of slurry soil evaporation were discussed. Fractal theory was used to evaluate the complexity of soil fissure under different setaria viridis roots, and the evaporation rate, fissure rate, average width of fissures and fissure expansion characteristics of the soil under different root content were analyzed. Finally, the mechanism of soil cracking under the action of soil evaporation with setaria viridis roots was discussed.

3. In the soil without setaria viridis roots, the sizes of the primary and secondary cracks formed by soil cracking were relatively uniform. When the soil contained setaria viridis roots, the crack width was large when the crack did not pass through the root system. When crossing the root system, the width of a single crack was small, the area of the formed cutting body was small, and the number was large. The root system controlled the crack expansion.

4. The fractal characteristics of the cracks with or without the root system were analyzed. In the process of crack propagation evolution, the fractal dimension changed as follows: the higher the root content, the smaller the fractal dimension. This also reflects the inhibitory effect of root systems on crack propagation.

5. In terms of the influence of the root system on the cracking rate of the soil, the higher the root system content, the lower the crack rate. At the same time, in terms of the average crack width, the higher the root system content, the smaller the average crack width. This also shows that, when the difference in the crack rate is small, the root system cuts the soil to a greater extent.

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